

Veronica Wang

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EDUCATION

Stanford University (B.S. Computer Science)

September 2025 - June 2029

Coursework: Machine Learning, Computer Vision, Applied Matrix Theory, Computer Organization & Systems, Linear Algebra & Multivariable Calculus, Trust and Safety, Probability for CS, Computer Graphics and Imaging

PUBLICATIONS

- Cohen, T.D., Oh, H. & Wang, V. Numerical study of computational cost of maintaining adiabaticity for long paths. Eur. Phys. J. A 62, 122 (2026). <https://doi.org/10.1140/epja/s10050-026-01879-4>

EXPERIENCE

Boeing | Software Engineering Intern

June-August 2026

- Implemented ramoops kernel crash persistence and designed a live serial logging system for the P-8 aircraft mission computer, allowing system states and kernel logs across 45 systems to be preserved for debugging.
- Worked with hardware to identify active console ports and evaluated a screen based approach to log each console session with tmux/telnet session management, systemd startup, and logrotate retention.

University of Maryland | Quantum Research Intern

May-August 2024

- Modeled random Hamiltonians and numerically solved the Schrödinger equation to compute a dimensionless adiabaticity-cost proxy, confirming its conjectured superlinear scaling relationship with path length.
- Built a simulation pipeline to orchestrate 1k+ runs with strict 10% error gates and validated results across 3 independent proxy formulations and multiple Hamiltonian draws to ensure the scaling behavior was robust.

PROJECTS

Transformer-based Neurosurgical Phase Classification

March-June 2026

- Designed a causal Transformer combined with Viterbi decoding and a learned transition matrix for surgical phase recognition on 40k aneurysm clipping frames with Stanford School of Medicine. Achieved 0.944 segmental F1 and ~95% validation accuracy, outperforming SOTA.
- Showed that fine-tuning ResNet50 on surgical frames (vs. frozen ImageNet weights) improved frame accuracy 53% to 95.7% and F1 0.553 to 0.895 with better tSNE clustering.

Illicit Bitcoin Wallet Classification

March-June 2026

- Proposed EvolveGCN-T, replacing GRU-based weight evolution with a Transformer self-attending over GCN weight matrix history. Outperformed EvolveGCN-O on Bitcoin-OTC edge classification (micro-F1 0.783 vs. 0.699, +8.4 pts) and reproduced published baselines to within +/- 1%.
- Designed a leakage-free train/test protocol for dynamic graph neural networks, improving robustness and generalization analysis for financial fraud detection.

JobShield: Full-Stack Fraudulent Job Posting Detection Platform

March-June 2026

- Built a three-tier fraud-detection platform ([Next.js](#), Supabase, Gemini 2.5 Flash) targeting active scam vectors. Achieved F1 0.913 and 95% recall at 4.5x lower inference cost than a pure-LLM (\$0.077/1k vs. \$0.35/1k).
- Logistic Regression + TF-IDF on postings handles obvious cases in under 0.05ms. Automatic feedback loop injects moderator decisions as few-shot examples for continuous improvement without retraining.

RFID-Based Assistive Audio Device at Chimes

October 2023-Jan 2024

- Designed an RFID-based audio identification device at the nonprofit Chimes to help custodial workers distinguish between cleaning supplies. Device to be scaled to hundreds of visually impaired employees.
- Built a two-ESP32 pipeline with RFID reader, ESP-NOW transmission, and SD card audio playback through a CAD-modeled enclosure. Blindfolded user study across 36 trials showed 30.1% faster item identification.

AWARDS

American Invitational Math Examination (AIME) Qualifier, USA Computing Olympiad (USACO) Gold Qualifier, NCWIT Aspirations in Computing Maryland Winner

SKILLS

Languages: Python, Java, C++, C, x86, Mathematica, LaTeX, NumPy, SciPy, Scikit-learn, PyTorch

Tools: Git, GCP, Vercel, Docker, Jira, Bitbucket, Unix, React, Next.js, Supabase, Onshape, Fusion 360, Blender

LEADERSHIP

Girls in Quantum | U.S. Outreach Lead

Dec 2023-May 2025

- Led global organization in 27 countries with 24k followers, managed website development and content strategy. Our group partnered with/was acknowledged by IBM, United Nations Girl Up, IEEE, and more.